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Lift3Dreamer: Boosting Text-Driven Novel View Synthesis via Lifted 3D Inpainting Model from Single Images.

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Manuscript Title Page

Title

Lift3Dreamer: Boosting Text-Driven Novel View Synthesis via Lifted 3D Inpainting Model from Single Images

Abstract

Text-guided novel view synthesis aims to generate controllable and semantically consistent images from a single input. However, existing methods often rely on pretrained diffusion models that lack geometric awareness, resulting in artifacts and inconsistencies in occluded or unobserved regions. In this work, we present Lift3Dreamer, a novel framework that fine-tunes a 2D inpainting diffusion model with pseudo-3D supervision derived from single-view RGB-D inputs. Specifically, we estimate depth from monocular images, lift them into 3D space, and simulate novel views via random camera motions. This process produces structured visibility masks that approximate real occlusions in 3D, which are then used to supervise the inpainting model with geometry-aware guidance. To scale training without paired data, we introduce a data-free pipeline combining large language models and text-to-image image generation. Equipped with these components, Lift3Dreamer shows strong performance in both synthetic and real-world scenarios, producing visually coherent and geometrically consistent results. Moreover, the framework can be extended to text-guided 3D-aware generation tasks, bridging the gap between 2D diffusion and view-consistent scene synthesis.

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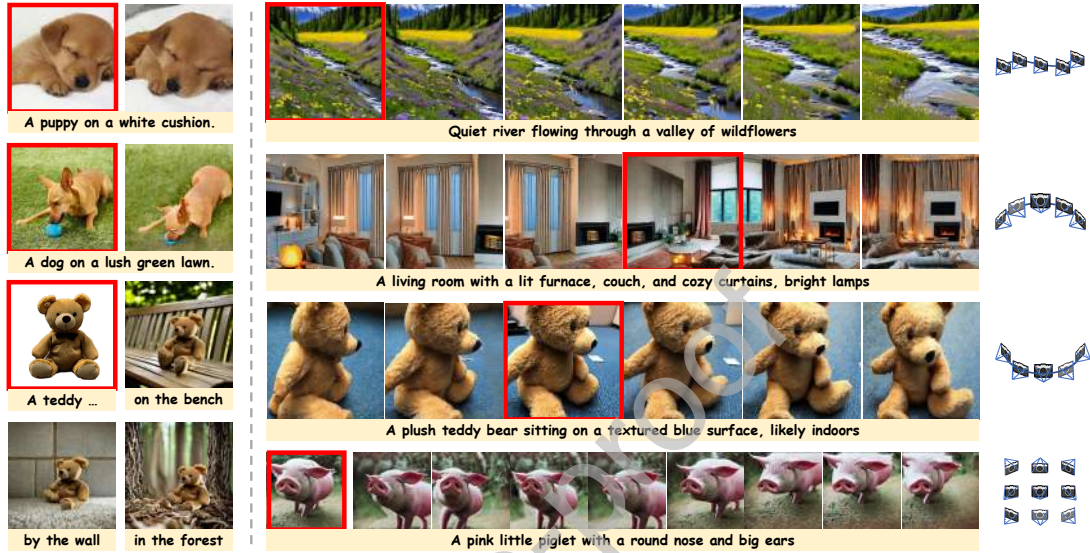
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Graphical Abstract

Lift3Dreamer: Boosting Text-Driven Novel View Synthesis via Lifted 3D Inpainting Model from Single Images

Highlights

- We propose Lift3Dreamer, a novel text-guided diffusion framework for controllable 3D scene generation without multi-view supervision.
- We introduce a data-free training pipeline using large language and diffusion models to synthesize high-quality RGB-D-text triplets. A 2D inpainting diffusion model is fine-tuned using pseudo-3D supervision lifted from monocular images and metric depth.
- Our method supports consistent novel view and multi-view synthesis guided by both geometry and text. Extensive experiments show state-of-the-art performance in rendering quality, geometric consistency, and text alignment.

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Lift3Dreamer

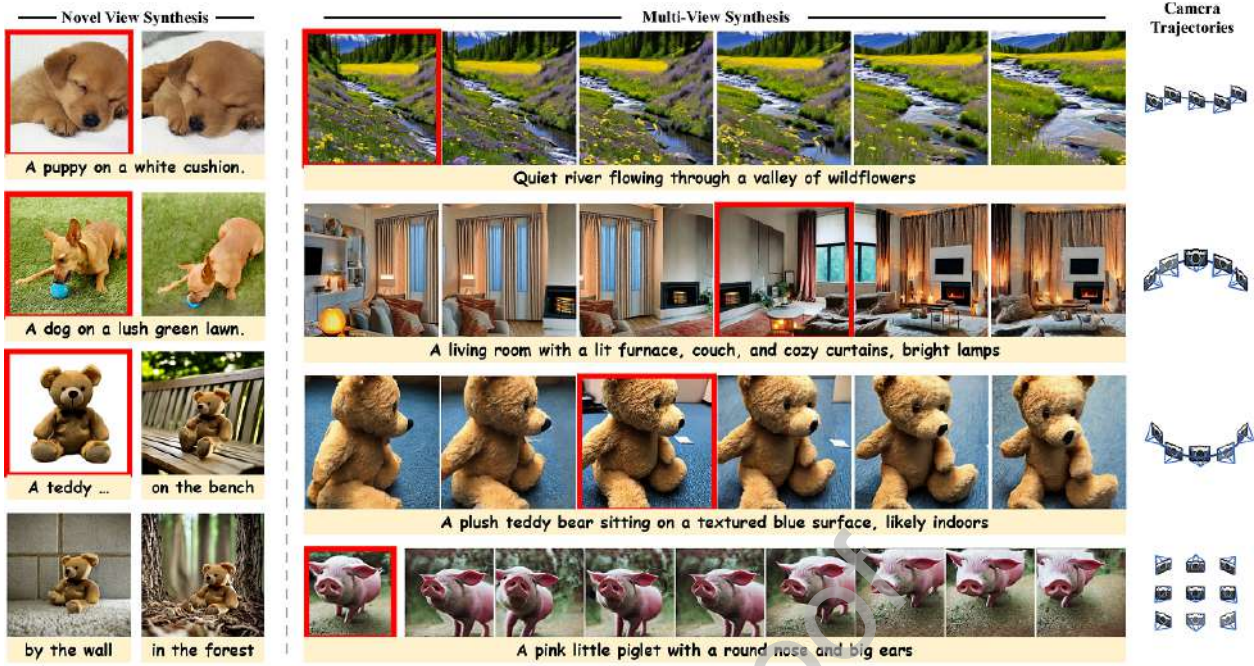


Figure 1: Generated 3D scenes from texts using our proposed 3DF-Dreamer. The left shows the results of text-guided novel view synthesis and editing with wide angles under 512×512 image resolution. The right shows the continuous and consistent scene synthesis with diverse camera trajectories.

1. Introduction

With recent advances in computer vision [12, 43, 62], text-guided novel view synthesis from a single image is enabling controllable and semantically consistent view generation under arbitrary camera trajectories. Applications span content creation [23], simulation [35], and virtual reality [33], where generating novel views from limited input is critical.

While recent advances in 2D diffusion models have improved image generation [42, 55], extending these capabilities to novel view synthesis remains challenging due to the lack of geometric awareness. Existing approaches, such as Score Distillation Sampling (SDS) [36, 8, 28], optimize 3D representations using pretrained diffusion priors, but are computationally expensive and typically limited to object-centric scenarios. More generalizable 3D diffusion models [29, 48] rely on large-scale multi-view datasets, which are difficult to collect for complex, diverse scenes.

To reduce reliance on 3D supervision, inpainting-based methods [17, 66, 19] have gained popularity. By hallucinating occluded regions using pretrained 2D inpainting models, these methods improve visual realism and inter-view consistency. However, most are trained with random or irregular masks [56, 32], which do not reflect the structured occlusions introduced by 3D warping. As a result, they often produce semantic artifacts or blurred textures under novel viewpoints, and struggle to maintain geometric consistency across views.

In this work, we propose **Lift3Dreamer**, a geometry-aware, text-guided inpainting framework tailored for high-quality novel view synthesis from a single image. Our key idea is to supervise a 2D inpainting diffusion model using pseudo-3D visibility masks derived from lifted RGB-D inputs. Specifically, we estimate depth from monocular images, lift the scene into 3D space, and simulate novel views via random camera transformations. Structured occlusions are then projected to create visibility masks, which guide the inpainting process.

To enhance realism and enable continuous view synthesis, we further introduce a mesh-based depth completion module, allowing us to recover dense geometry in previously hallucinated regions. Once fine-tuned, the inpainting model can generate novel views that are both semantically faithful and geometrically aligned with the input, under user-provided text prompts.

To enable scalable training, we also design a data-free pipeline that constructs large-scale pseudo-paired training samples. We use a large language model (e.g., GPT-4) to generate diverse textual prompts, synthesize corresponding high-resolution images via Stable Diffusion XL [55], and estimate monocular depth using ZoeDepth [2]. These are then warped into new views to simulate occlusions and form supervision signals.

Extensive experiments on both synthetic and real-world data demonstrate that Lift3Dreamer achieves superior rendering quality, view consistency, and controllability. As shown in Figure 1, our model can synthesize photorealistic and geometrically coherent novel views, even from a single input image, supporting flexible camera control and prompt-based scene

variation. Our main contributions are as follows:

- We propose **Lift3Dreamer**, a novel text-guided diffusion framework for view-consistent novel image synthesis using pseudo-3D supervision from single-view RGB-D inputs.
- We fine-tune a 2D inpainting diffusion model using geometry-aligned masks derived from simulated views and completed depths, enabling semantically consistent and 3D-aware novel view synthesis.
- We introduce a scalable, data-free training pipeline that generates high-quality image-text-depth triplets using large pre-trained models without requiring multi-view or real 3D data.

2. Related Work

Text-to-3D generation. Early approaches predominantly rely on Generative Adversarial Networks (GANs) [6, 7]. In parallel, some methods adopt alternative 3D representations, such as Neural Radiance Fields (NeRF) [14, 49, 22], and their derivatives. These approaches typically involve training a 3D scene generator guided by a jointly optimized discriminator [15, 57, 24], enabling photorealistic 3D content generation. However, a major limitation of these methods is their reliance on calibrated camera poses or accurate pose estimation.

To mitigate this dependency, DreamFusion [36] proposed Score Distillation Sampling (SDS), a novel technique that leverages a 2D diffusion model as a prior to supervise NeRF optimization from text prompts. Subsequent methods, including Magic3D [28], ProlificDreamer [59], and Fantasia3D [8], have improved output quality, resolution, and diversity. However, these optimization-based pipelines are computationally expensive and typically limited to single-object or scene-specific generation.

Recent works [29, 47, 48, 30, 31, 20] aim to overcome these constraints by developing generalizable 3D-aware diffusion models trained on large-scale datasets such as Objaverse [10]. These methods adopt an end-to-end paradigm to learn 3D representations from single-view or weakly-supervised data. But achieving high-fidelity geometry remains a challenge, due to the scarcity of high-quality 3D annotations or multi-view images.

More recently, methods such as Text2NeRF combine NeRF with text-to-image diffusion priors to construct content and geometric cues, using progressive inpainting to enforce multi-view consistency [63]. RealmDreamer extends Gaussian splatting with inpainting and depth-conditioned diffusion, enabling high-quality, forward-facing scene synthesis across diverse prompts without multi-view data [50]. However, RealmDreamer relies heavily on pretrained 2D inpainting models that are trained on random or unstructured masks, leading to suboptimal performance when dealing with structured occlusions caused by 3D warping. This results in artifacts or blurred regions, especially under novel views, due to the lack of explicit geometric supervision or alignment across views.

In our approach, we address these limitations by leveraging large text-to-image diffusion models in conjunction with monocular depth estimation. By aligning the inpainting training with the geometry of view synthesis, our method enables semantically consistent and geometry-aware novel view generation from a single image.

Text-guided diffusion models. Text-guided models show superior ability to synthesize high-fidelity images from texts [44, 39, 38]. These models are rooted in the framework of denoising diffusion probabilistic models (DDPMs) [16], which iteratively convert random noise into samples from a target distribution through a Markov chain. The Latent Diffusion Models (LDM) [42] perform the diffusion steps in the latent image space, which reduces the computation cost. Text-to-image diffusion models achieve state-of-the-art image synthesis results by encoding text inputs into latent vectors via pre-trained language models like CLIP [37]. Therefore, LDM and text encoding enable high-quality text-to-image synthesis [53] and text-guided masked-image inpainting [54]. We utilize such text-guided diffusion models as 2D priors to paint detailed textures that are coherent and consistent across different views. However, natively applying such inpainting models fails to generalize well to the masks that arise from the warping procedure since they are trained with random masks. Therefore, we adopt a warp-back strategy with monocular depth estimation to generate realistic masks for training via camera motions.

Monocular depth estimation. Monocular depth estimation (MDE) is a pivotal task in computer vision. It is essential for many vision applications [11, 18, 26], aiming to deduce the depth information of a scene from a single-view image. The prior progression in using convolutional neural networks (CNNs), recurrent convolutional neural networks (RNNs), and Transformers yields a considerable improvement [21, 1, 13, 40]. MiDaS [41] develops tools that enable mixing multiple datasets during training, even if their annotations are incompatible, and further proposes a robust training objective that is invariant to changes in depth range and scale. This advocates using principled multi-objective learning to combine data from different sources, leading to a model with excellent generalization performance. Based on MiDaS, recent ZoeDepth [2] focuses on both generalization performance on relative depth estimation and metric depth estimation, leading to a model with excellent generalization performance while maintaining metric scale. Therefore, we turn to monocular depth estimation to find the bridge between single-view images and 3D worlds.

Lift3Dreamer

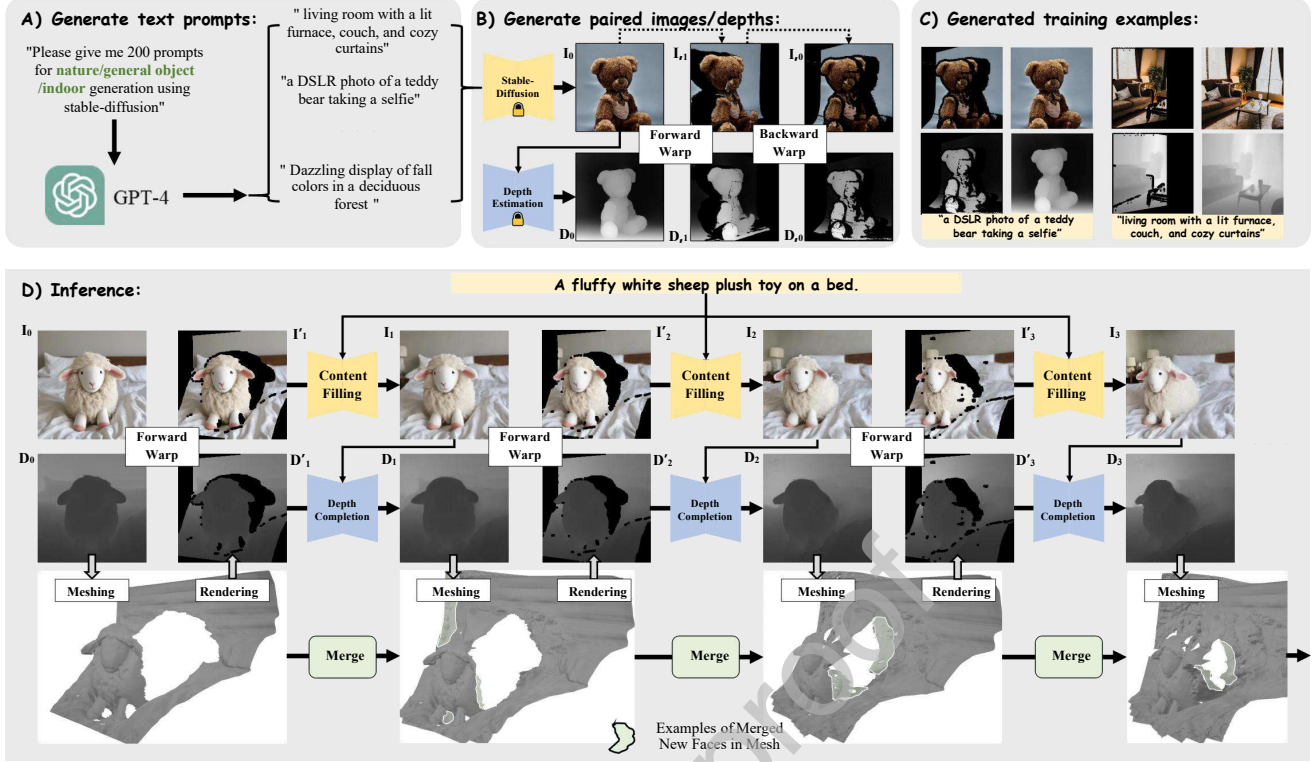


Figure 2: Illustration of our proposed training data generation and 3D-aware image inference pipeline.

3. Methods

3.1. Formulation

Motivated by the limitations of existing text-to-3D generation approaches—particularly their lack of geometric consistency due to training with random masks—we propose Lift3Dreamer, a geometry-aware framework that simulates 3D supervision without requiring multi-view data. Our method leverages monocular depth estimation and camera simulation to generate pseudo-3D occlusions, which are used to fine-tune a 2D text-guided inpainting diffusion model.

We decompose the novel view synthesis task into two key steps: 1) simulating structured occlusions via forward warping, and 2) inpainting the occluded content in a geometry-aligned manner.

Given a single-view RGB image I_0 , we first estimate its monocular depth D_0 using a pretrained depth model Φ :

$$D_0 = \Phi(I_0). \quad (1)$$

We lift the RGB-D input into 3D space using the intrinsics matrix $\mathbf{K}$ and simulate a novel viewpoint via a transformation matrix $\mathbf{T}_{0 \rightarrow 1}$ randomly sampled to reflect natural camera motion. For each pixel $\mathbf{p}_0$ in the source image, its projected coordinate in the novel view $\mathbf{p}_1$ is computed as:

$$\mathbf{p}_1 \sim \mathbf{K} \mathbf{T}_{0 \rightarrow 1} \mathbf{D}_0(\mathbf{p}_0) \mathbf{K}^{-1} \mathbf{p}_0. \quad (2)$$

Using forward warping, we render a novel view image I'_1 that only contains pixels visible from the source view. This inherently creates a structured hole mask $\mathbf{H}$ for pixels not covered by any source projection—representing true occlusions that arise in the 3D view synthesis process.

To recover the full novel view I_1 , we train a content inpainting diffusion model Ψ conditioned on I'_1 and the binary mask $\mathbf{H}$:

$$I_1 = \Psi(I'_1, \mathbf{H}). \quad (3)$$

Unlike conventional inpainting methods trained with random or unstructured masks, our method uses these pseudo-3D structured masks that simulate real occlusion patterns. This enables the model to learn to hallucinate geometry-consistent content aligned with realistic camera motion.

Due to the lack of paired ground truth novel views, we introduce a data-free training strategy described in Section 3.2, which uses synthetic RGB-D images and text prompts to generate supervision signals. Furthermore, we extend this formulation to multi-view synthesis with a mesh completion pipeline described in Section 3.4, enabling continuous and consistent camera trajectory rendering.

3.2. Data-Free Training Process

Obtaining ground-truth novel views with corresponding masks typically requires access to dense 3D datasets, which are scarce and expensive to acquire. To overcome this limitation, we propose a data-free pipeline that synthesizes pseudo-3D training pairs from single-view RGB images paired with estimated depth.

Our core idea is to simulate the occlusion patterns that naturally occur in 3D view synthesis using a warp-and-warp-back strategy. As shown in Figure 2B, given a synthesized image $\mathbf{I}_0$ and its estimated depth $\mathbf{D}_0$, we apply forward warping to obtain a partial novel view $\mathbf{I}'_1$ under a random camera transformation $\mathbf{T}_{0 \rightarrow 1}$. This generates a visibility-aware mask $\mathbf{H}_1$, where unobserved pixels (i.e., occlusions) are labeled as holes.

To train the inpainting diffusion model Ψ , we need to provide it with input-target pairs that reflect realistic 3D occlusion conditions. However, we lack the true novel view image $\mathbf{I}_1$. Therefore, we perform backward warping from $\mathbf{I}'_1$ to reconstruct a degraded source view $\mathbf{I}'_0$, using the inverse transform $\mathbf{T}_{1 \rightarrow 0}$:

$$\mathbf{p}'_0 \sim \mathbf{K}^{-1} \mathbf{T}_{1 \rightarrow 0} \mathbf{D}'_1(\mathbf{p}'_1) \mathbf{K} \mathbf{p}'_1, \quad (4)$$

where $\mathbf{p}'_1$ denotes visible pixels in $\mathbf{I}'_1$. The resulting $\mathbf{I}'_0$ now contains realistic holes $\mathbf{H}_0$ induced by the simulated 3D transformation. We then train Ψ to reconstruct the original source image $\mathbf{I}_0$:

$$\mathbf{I}_0 = \Psi(\mathbf{I}'_0, \mathbf{H}_0). \quad (5)$$

This effectively enables pseudo-paired training: the visible region of $\mathbf{I}'_1$ is warped back to produce $\mathbf{I}'_0$ (with missing regions), and $\mathbf{I}_0$ serves as ground truth. Importantly, the hole masks $\mathbf{H}_0$ accurately reflect structured visibility loss due to viewpoint change, rather than random erasure.

To scale this training pipeline, we generate synthetic RGB-D-text triplets. We use GPT-4 [34] to produce a diverse set of scene descriptions $\mathbf{T}$, which are rendered into high-resolution images using Stable-Diffusion-XL [55]. We estimate monocular metric depth for each generated image using ZoeDepth [2]. These RGB-D-text samples serve as the source data for warping-based training pair generation.

To incorporate text guidance during inpainting, we extend the model Ψ to accept prompt $\mathbf{t}$ as input, enabling the network to leverage both geometric and semantic cues:

$$\mathbf{I}_0 = \Psi(\mathbf{I}'_0, \mathbf{H}_0, \mathbf{t}). \quad (6)$$

Rather than training from scratch, which can lead to poor visual fidelity, we fine-tune a high-quality pretrained text-to-image diffusion model using our pseudo-paired data. This ensures both semantic plausibility and geometric consistency in novel view hallucination.

3.3. Training Text-Guided Diffusions

We fine-tune a text-guided inpainting diffusion model Ψ using the pseudo-paired data generated via our warp-back pipeline. These training samples consist of masked images $\mathbf{I}'_0$ with structured holes $\mathbf{H}_0$ that simulate real 3D occlusions, paired with original images $\mathbf{I}_0$ and their corresponding text prompts $\mathbf{t}$. This setup enables the model to learn how to hallucinate semantically plausible and geometrically consistent content guided by both the input image and text.

Our implementation is based on the Stable Diffusion framework [53], a state-of-the-art latent text-to-image diffusion model. Stable Diffusion operates in the latent space of a pretrained variational autoencoder (VAE), composed of an encoder $\mathcal{E}$ and decoder $\mathcal{D}$. The input image $\mathbf{I}'_0$ is first encoded into a latent vector $\mathbf{z} = \mathcal{E}(\mathbf{I}'_0)$, which is progressively noised into $\mathbf{z}_t$ across t time steps. The model Θ is trained to predict the added noise ϵ at each step, conditioned on the masked latent, the original (unmasked) latent, and the text prompt:

$$L = \mathbb{E}_{\mathbf{z}, \mathbf{z}_t, \mathbf{t}, \epsilon \sim \mathcal{N}(0,1), t} \left[\left\| \epsilon - \epsilon_\theta(\mathbf{z}_t, t, \mathcal{E}(\mathbf{I}'_0), \mathbf{t}) \right\|_2^2 \right]. \quad (7)$$

To support masked image conditioning, we follow InstructPix2Pix [3] and modify the input layers to incorporate additional channels for the binary mask $\mathbf{H}_0$ and the original latent $\mathcal{E}(\mathbf{I}'_0)$. The additional layers are initialized to zero, while the rest of the network is initialized from a pretrained Stable Diffusion checkpoint.

This fine-tuning approach enables Ψ to restore occluded content that is not only semantically aligned with the prompt but also geometrically plausible given the masked image and its visibility pattern. An example of such inpainting behavior is shown in Figure 4.

Lift3Dreamer

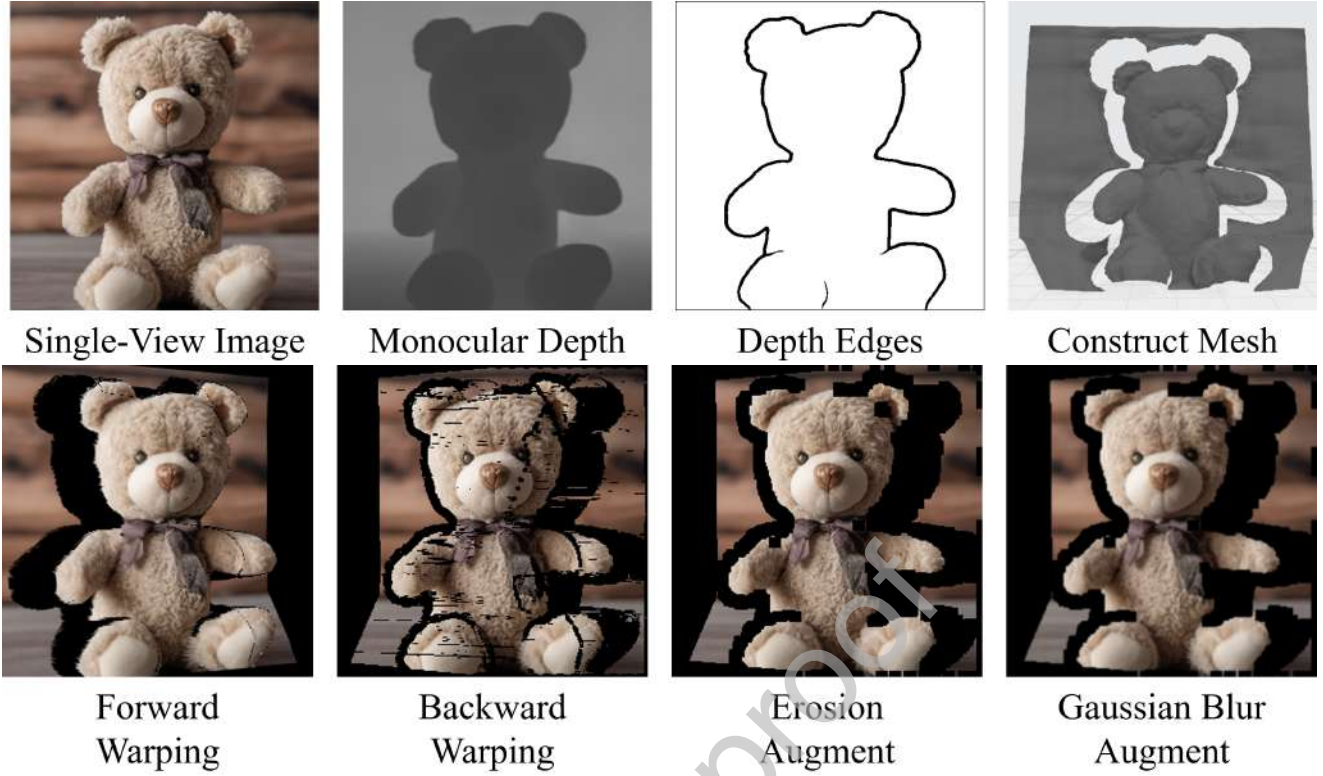


Figure 3: Illustration of the mesh construction pipeline for inpainting data generation for training.

3.4. Mesh Completion for Multi-View Synthesis

To enable continuous and consistent novel view synthesis along arbitrary camera trajectories, it is essential to maintain depth completeness across frames—particularly for newly hallucinated regions that were not visible in the original input.

As illustrated in Figure 2B and D, the first novel view I_1 is synthesized by warping the source view I_0 to obtain a partial view I'_1 and inpainting the missing content via our fine-tuned model Ψ :

$$I_1 = \Psi(I'_1, H_1, t). \quad (8)$$

To synthesize the next view I_2 , both I_0 and I_1 provide contributions. However, some pixels in I_1 represent newly hallucinated content not present in I_0 , and thus lack depth information. These RGB pixels are generated purely by Ψ and are not associated with any projected 3D geometry.

To address this, we introduce a depth completion network Ω , which estimates the missing depth values for inpainted regions using image guidance. Given a partial depth map D'_1 obtained by forward warping and the corresponding inpainted image I_1 , we complete the depth as:

$$D_1 = \Omega(D'_1, H_1, I_1). \quad (9)$$

With I_1 and D_1 , we lift pixels to 3D space and construct an incremental mesh by connecting neighboring pixels into triangle faces based on their 2D grid topology. To avoid erroneous connections, we discard long edges by thresholding depth discontinuities, following AdaMPI. This mesh is then rendered into the next target view to obtain both the warped image and its corresponding depth map.

By repeating this process iteratively across frames, we generate a coherent multi-view sequence with consistent geometry, enabling high-quality text-driven 3D scene synthesis from a single input image.

4. Experiments

We evaluate the performance of **Lift3Dreamer** on text-guided novel view and multi-view synthesis tasks. This section first outlines the implementation details of our system. We then present both qualitative and quantitative comparisons with state-of-the-art methods across a range of benchmarks. Finally, we demonstrate the versatility of Lift3Dreamer under various input conditions and continuous camera trajectories.

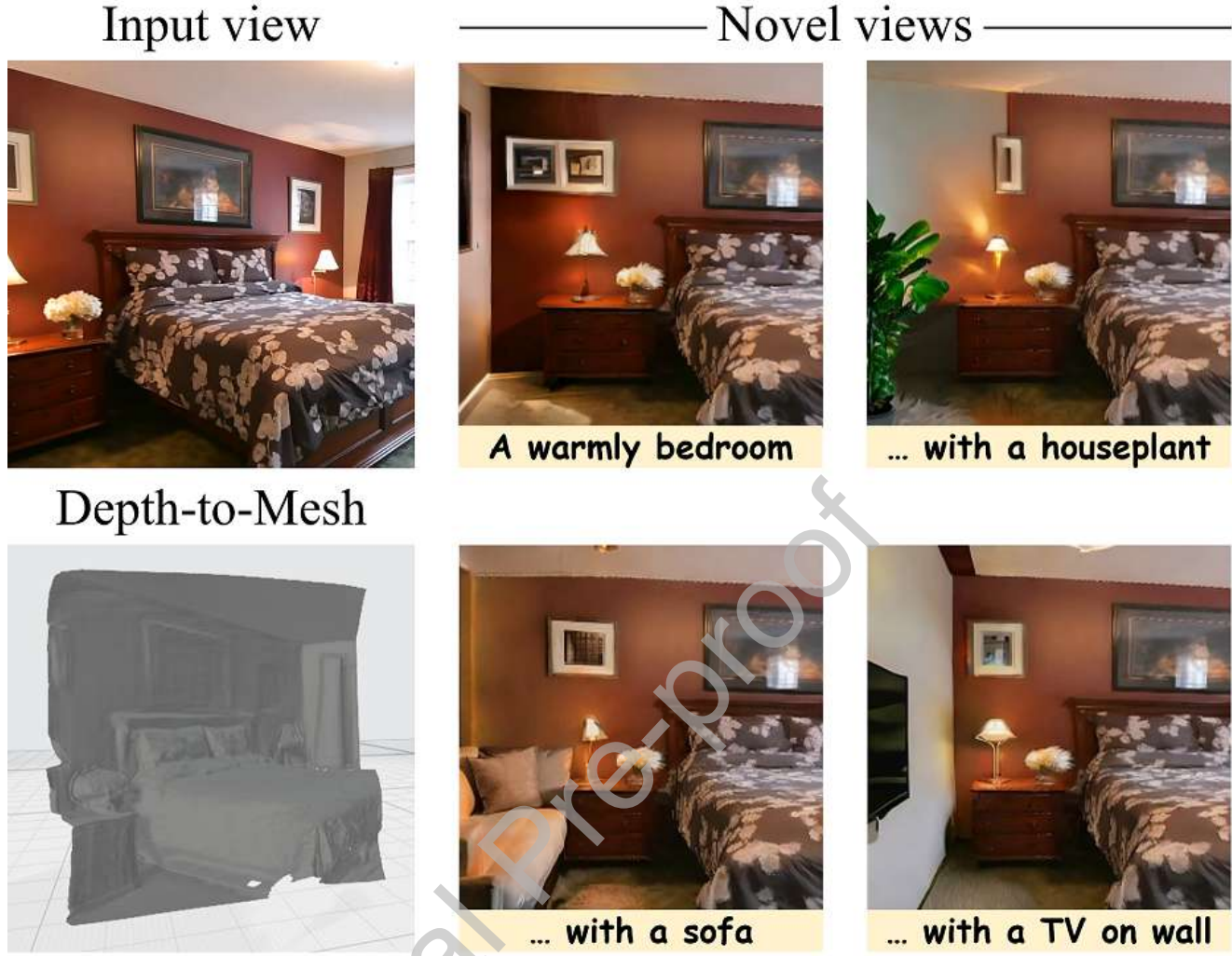


Figure 4: Controllable novel-view scene synthesis with user-given text guidance to edit the content.

Extensive results confirm that Lift3Dreamer produces semantically consistent and geometrically coherent views, even from a single input image. Our method generalizes well to diverse styles and scenes, showcasing strong performance across both synthetic and real-world scenarios.

4.1. Implementation Details

Training Data. To train our inpainting model, we construct a large-scale synthetic dataset using diverse prompt sets covering general objects, natural environments, and indoor scenes. In total, we generate 200,000 images from 2,000 prompts using Stable Diffusion XL [55] at a resolution of 1024×1024 . These images exhibit complex geometric structures and diverse appearances, making the resulting dataset well-suited for training 3D-aware image synthesis models. Corresponding monocular depth maps are estimated using ZoeDepth [2], which demonstrates strong generalization to synthetic content.

Training Settings. We assume a default normalized camera intrinsics setting with a relative focal length of 1.0. The forward and backward warping operations may introduce blur due to interpolation artifacts and mesh rasterization. To address this, we apply random Gaussian blur augmentation to the warp-back images during training, following IViD [60]. This helps bridge the distribution gap between synthetic and real-world view synthesis conditions.

We train our model at a target resolution of 512×512 using the same network architecture and optimization settings as Stable Diffusion 2.0 Inpainting. Training is conducted on 8 NVIDIA Tesla V100 GPUs (32GB each) for 100,000 steps over our synthetic dataset.

Lift3Dreamer

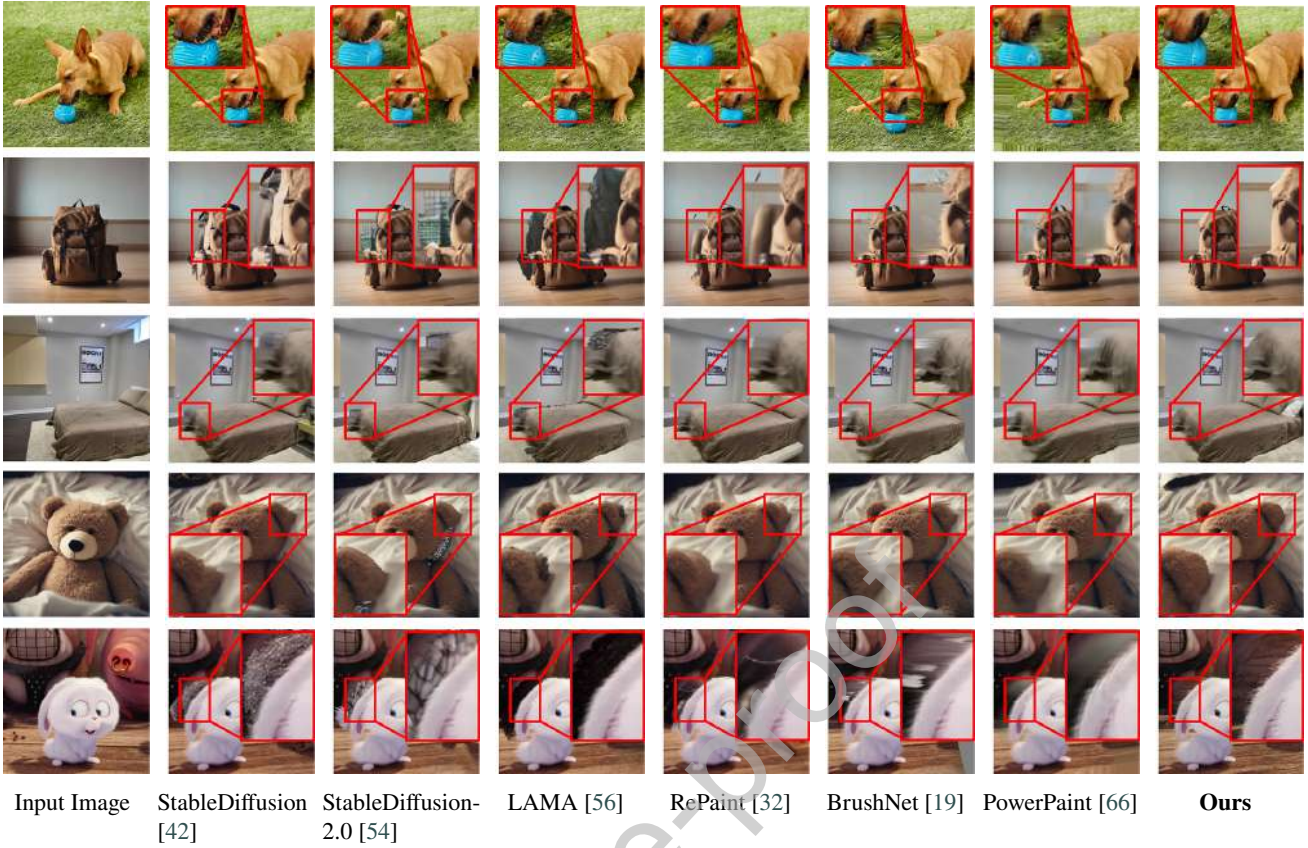


Figure 5: Quantitative comparison with state-of-the-art inpainting methods on novel view synthesis from single-view images.

Method	LPIPS ↓	PSNR ↑	SSIM ↑
SVMPI [57]	0.162	23.68	0.799
MINE [22]	0.169	23.45	0.802
3D-Photo [49]	0.178	21.72	0.730
AdaMPI [15]	0.145	23.76	0.802
StableDiffusion [42]	0.155	22.17	0.762
StableDiffusion-2.0 [54]	0.160	22.01	0.775
LAMA [56]	0.136	23.71	0.794
RePaint [32]	0.132	24.02	0.820
BrushNet [19]	0.135	23.96	0.813
PowerPaint [66]	0.131	24.19	0.826
Our	0.129	24.41	0.840

Table 1

Quantitative comparison with advanced inpainting models on RealEstate-10K dataset.

4.2. Comparison with State-of-the-art Methods

Compared Methods. We evaluate our method on novel view synthesis by comparing it to prior depth-based approaches with 3D-Photo [49] and three MPI-based methods, including SVMPI [57], MINE [22], and AdaMPI [15]. We also compare Lift3Dreamer with other state-of-the-art inpainting models by replacing the trained content inpainting model, including Stable-Diffusion-Inpainting (SD) [42], Stable-Diffusion-Inpainting 2.0 (SD-2.0) [54], LAMA [56], RePaint [32], BrushNet [19], and PowerPaint [66]. SD [42] and SD-2.0 [54] are both text-guided inpainting models based on text-guided diffusions.

Evaluation dataset. we use the RealEstate-10K dataset [65] for evaluation, which provides multi-view images of static scenes. We sample 800 clips provided by AdaMPI from the test set for evaluation. We specify a randomly sampled frame for each clip as the source view and the following 10th frame as the target view. For fair comparison, we use DPSNet [64] and COLMAP [46] to estimate the depth and the camera parameters as AdaMPI does. We select image pairs with moderate camera motion for evaluation.

Quantitative Results. To ensure fair comparison, we use the same depth maps and camera parameters for all methods. We

Lift3Dreamer

Method	CLIP $\uparrow$	Depth Pearson $\uparrow$	IS $\uparrow$
Text2Room [17]	28.11	0.77	5.10
DreamFusion [36]	29.48	0.09	6.80
ProlificDreamer [59]	29.39	0.16	6.89
LucidDreamer [27]	29.97	0.80	5.73
RealmDreamer [50]	31.69	0.89	6.99
Ours	32.02	0.93	7.08

Table 2

Quantitative results for the Text-to-3D generation task, including CLIP alignment and other metrics to evaluate 3D scene quality and rendering fidelity.



Figure 6: Qualitative results on Text-to-3D generation task with RealmDreamer. We show the inpainting results of novel views (partial) using Stable-Diffusion 2.0 with and without our proposed Lift3Dreamer for fine-tuning.

report the results of MINE implemented by [15] as official MINE [22] does not support depth input. We also compare our method to some state-of-the-art inpainting by replacing our trained content inpainting model with them. To enable text-guided inpainting for novel view synthesis using text-driven inpainting models, i.e., SD [42], SD-2.0 [54], and our proposed Lift3Dreamer, we utilize captions predicted by GenerativeImage2Text model [58] as the input text prompts. We report the SSIM, PSNR, and LPIPS scores to measure the rendering quality. As illustrated in Table 1, our method demonstrably surpasses others in performance.

Qualitative Results. Figure 5 displays the qualitative results of our method against existing ones. Uniform predicted depth maps from ZoeDepth [2] are used for models that incorporate depth inputs. The qualitative analysis underscores that Lift3Dreamer substantially reduces blurring and artifacts typically engendered by 3D motion and inpainting, maintaining reasonable detail. Moreover, Lift3Dreamer demonstrates robust generalizability across a wide array of scenes and objects. In subsequent sections, we will delve deeper into additional results showcasing the capabilities of Lift3Dreamer in novel view and multi-view synthesis.

4.3. Extension to Text-to-3D Generation

While Lift3Dreamer is designed for novel view synthesis, it can be extended to support text-to-3D scene generation tasks that rely on consistent multi-view rendering. In particular, we integrate our fine-tuned inpainting model into the RealmDreamer [50] pipeline by simply replacing its default Stable Diffusion 2.0 Inpainting model. This substitution enables geometry-aware faithful view completion within the text-to-3D reconstruction process.

Compared Methods. We compare our integrated model against several state-of-the-art text-to-3D methods, including Text2Room [17], DreamFusion [36], ProlificDreamer [59], LucidDreamer [27], and RealmDreamer [50]. Among these, diffusion-based models like ProlificDreamer and DreamFusion often struggle with geometric fidelity or prompt adherence, while inpainting-based methods such as Text2Room lack generalizability beyond room environments. RealmDreamer pro-

Lift3Dreamer

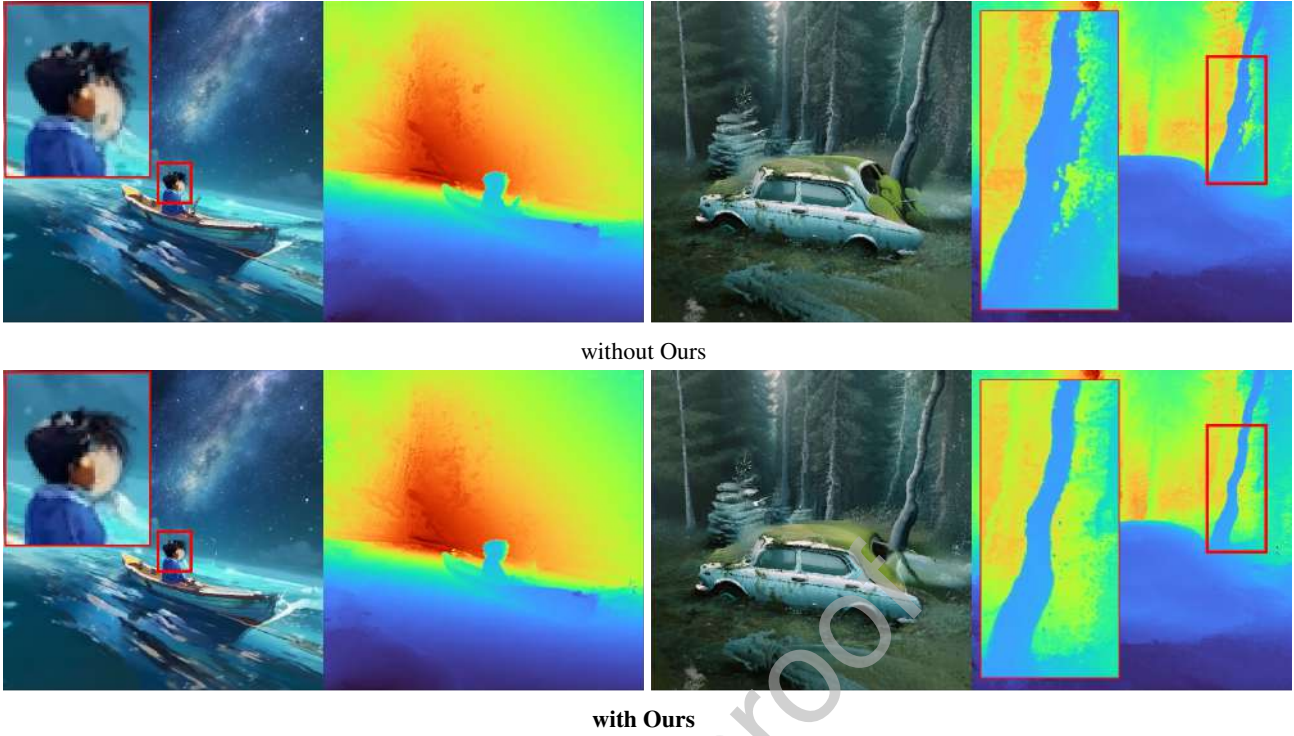


Figure 7: Qualitative results on Text-to-3D generation task with RealmDreamer using Stable-Diffusion 2.0 with and without our proposed Lift3Dreamer, where we show both the rendered images and depths.

vides a strong baseline by combining inpainting and sparse geometry lifting, but its inpainting model is not fine-tuned for 3D occlusion structures, leading to degraded quality in novel views.

Quantitative Results. Tab. 2 reports results on three key metrics: CLIP score [37] for text-image alignment, Inception Score (IS) [45] for visual fidelity, and Pearson correlation between rendered and predicted depth (from DepthAnythingV2 [61]) for geometric consistency. Our approach improves over the original RealmDreamer on all metrics, highlighting the effectiveness of our inpainting fine-tuning in producing geometrically aligned renderings.

Qualitative Results. Figs. 6 and 7 illustrate qualitative improvements by replacing RealmDreamer’s inpainting module with our Lift3Dreamer. The results demonstrate sharper textures, better structure preservation, and improved depth consistency across novel views, especially in challenging occluded or extrapolated regions.

4.4. Extensions to Continuous View Synthesis

We further evaluate the ability of Lift3Dreamer to generate continuous and view-consistent images under diverse camera trajectories. This experiment highlights our method’s capability for 3D-aware image synthesis with free-viewpoint control. Additional examples are provided in the supplementary material.

3D-Aware Image Synthesis. We perform qualitative comparisons on a 3×3 viewpoint grid centered around the canonical view. Our method is compared against representative 3D-aware image generation baselines, including GAN-based models (pi-GAN [7], EpiGRAF [52], and EG3D [6]) and the recent diffusion-based model IVID [60]. These models are trained to produce class-conditioned images.

For fair comparison, we generate images using the prompt template: “A DSLR photo of a [C]”, where “[C]” denotes a specific object category. Camera poses are randomly sampled from Gaussian distributions with standard deviations. Our method synthesizes each novel view independently using the fine-tuned inpainting model guided by pseudo-3D warping and text prompts. Figure 8 illustrates the qualitative results. Compared to existing methods, Lift3Dreamer produces images that exhibit significantly better view consistency, geometry-aware content extrapolation, and semantic fidelity. Moreover, unlike class-conditioned GANs, our method supports both arbitrary textual descriptions and free-viewpoint synthesis, enabling flexible scene control beyond real-world class constraints.

Perpetual View Synthesis. This task aims to creating unbounded flying-through videos of scenes, starting from a single view. This task typically involves extended linear camera movements. We compare our Lift3Dreamer with other notable methods in this arena, including Diffdreamer [4], SceneDreamer [9], InfiniteNature-Zero [25], and Persistent Nature [5]. These methods have shown impressive results in generating long-range views and are generally trained on the expansive LHQ dataset [51], encompassing single-view images of natural landscapes. The qualitative results in Figure 9 demonstrate

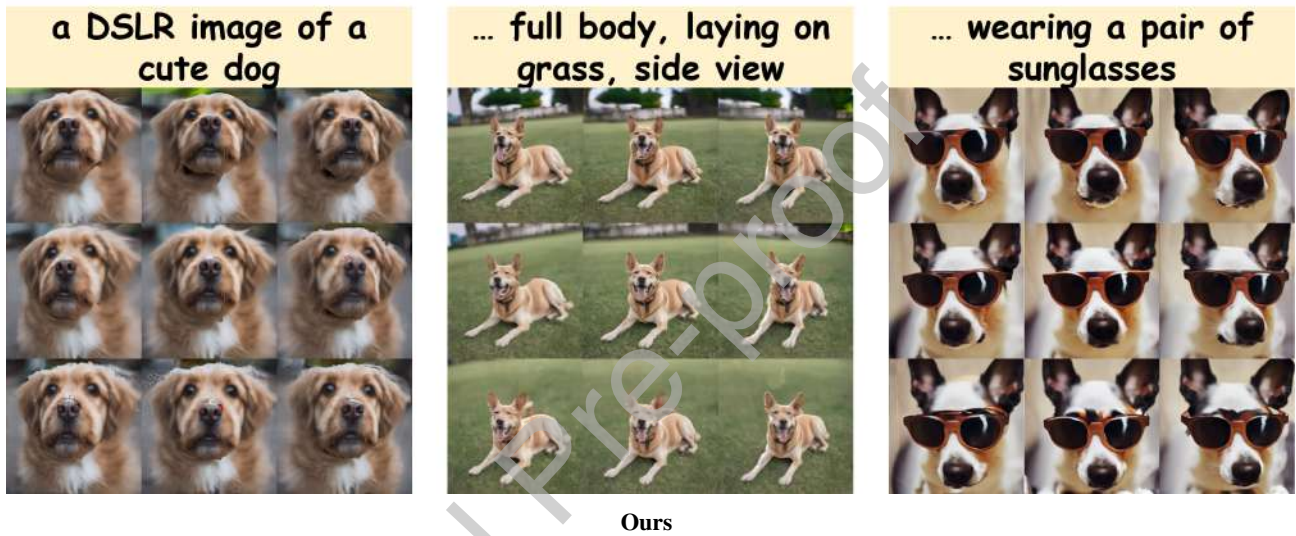
Lift3Dreamer



EpiGRAF [52]

EG3D [6]

IVID [60]



Ours

Figure 8: Qualitative results on 3d-aware image synthesis. Previous methods perform unconditional synthesis on a fixed dataset. Our proposed method performs better visual quality with class prompts. Besides, our method enables controllable synthesis with user-given text for guidance.

the remarkable capabilities of Lift3Dreamer in perpetual view synthesis, yielding competitive results without necessitating training or fine-tuning on real landscape images. Moreover, Lift3Dreamer’s ability to adhere to user-defined text prompts allows for precise synthesis of content that aligns with user expectations.

5. Conclusion

In this paper, we present **Lift3Dreamer**, a geometry-aware and text-guided diffusion framework for novel view synthesis from a single image. By leveraging monocular depth estimation and simulating 3D camera motion, we construct structured visibility masks that reflect real occlusions, enabling the fine-tuning of a 2D inpainting diffusion model with geometry-aligned supervision. To facilitate scalable training, we propose a data-free pipeline that synthesizes large amounts of pseudo-paired data using large language models and text-to-image generation. In addition, we introduce a mesh-based depth completion module to support continuous view synthesis along arbitrary camera trajectories, enhancing consistency across synthesized frames. Lift3Dreamer bridges the gap between 2D diffusion models and geometry-consistent novel view generation. It supports semantically controllable, high-quality rendering from a single image and can be extended to broader 3D-aware generation tasks, offering a scalable path toward text-driven 3D understanding from minimal supervision.

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Lift3Dreamer



Diffdreamer [4]



SceneDreamer [9]



InfiniteNature-Zero [25]



Persistent Nature [5]



A DSLR photo of windy coast with tall cliffs and waves



A desolate stretch of highway in the west, resembling an apocalyptic wasteland, winds through a barren desert



A snow-covered mountains under a clear blue sky with white clouds

Ours

Figure 9: Qualitative results on perpetual view synthesis.

Declaration of Interest Statement

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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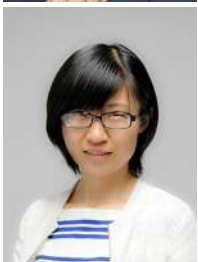
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